

Supplementary Material for

Bridged Semantic Alignment for Zero-shot 3D Medical Image Diagnosis

Appendix A. GPT-4 Turbo Annotation and Expert Verification

We employed GPT-4 Turbo to automatically label long-tail abnormalities in the CT-RATE-LT dataset using the prompt illustrated in Fig. S1. This prompt was designed to guide the model in extracting diagnostic findings from radiology reports and assigning binary presence/absence labels for each abnormality category.

After the automatic labeling stage, all GPT-4 Turbo-predicted abnormality labels were reviewed by a **board-certified radiologist with over ten years of diagnostic experience**. Instead of re-annotating the reports from scratch, the radiologist performed a **targeted verification** process focusing on the correctness and completeness of the model-generated labels.

Specifically:

- **For positive labels (predicted as 1):** Each case was checked to confirm whether the report text explicitly or implicitly supported the abnormality. False positives (e.g., negated findings misclassified as positive) were corrected to 0.
- **For negative labels (predicted as 0):** We used a keyword-based pre-screening to efficiently surface suspected false negatives. For each category, we compiled a curated lexicon (including common synonyms and abbreviations). If the report text contained any lexicon term for that category while GPT-4 Turbo predicted 0, the case was automatically flagged for expert review (explicit negations such as “no evidence of . . .” were not flagged). The radiologist then verified these flagged cases; confirmed misses were corrected from 0 to 1, thereby improving recall.

To ensure transparency and quantitative reproducibility, we measured multiple verification and agreement statistics between the GPT-4 Turbo-generated labels and the expert-corrected labels. Table I summarizes the statistical results.

These results demonstrate that GPT-4 Turbo achieved high consistency with expert judgment, with a Cohen’s κ of 0.91 indicating *almost perfect* agreement, and an overall accuracy of 97.9%. The small correction rates (2.1% label-level) confirm that only minor expert adjustments were necessary, validating the reliability and scalability of the automated annotation pipeline.

In addition, the expert analysis revealed that GPT-4 Turbo’s most frequent errors occurred in cases with subtle or overlapping descriptions, such as:

- Missing early-stage *pulmonary fibrosis* when *pneumonia* was labeled positive, leading to underdiagnosis of coexisting fibrotic changes.
- Overinterpreting benign pulmonary nodules as *lung masses*.
- Frequent missed diagnoses of *spondylosis* were observed, likely due to its subtle radiologic manifestations and peripheral mention in reports, where age-related degenerative changes are often described in mild or incidental terms.

Representative examples of corrections (original report → GPT-4 Turbo output → expert-corrected label) are provided in Table II.

Table I: Quantitative verification statistics of GPT-4 Turbo labeling and expert correction.

Metric	Definition	Value
Total unique annotated cases	Number of unique cases included for expert verification	1,109
Total abnormality categories	Number of disease categories annotated per case	15
Average false-positive rate (misdiagnosis)	Proportion of GPT-4 Turbo labels incorrectly predicted as positive	1.44%
Average false-negative rate (missed diagnosis)	Proportion of GPT-4 Turbo labels incorrectly predicted as negative	0.66%
Label-level correction rate	Fraction of all abnormality labels revised during expert verification	2.10%
Cohen’s κ (GPT-4 Turbo vs. expert)	Agreement between GPT-4 Turbo and expert labels, adjusted for chance	0.91
Overall percent agreement	Exact match rate between GPT-4 Turbo and expert labels	97.9%

Table II: Illustrative examples of GPT-4 Turbo labeling errors and expert corrections.

Category	Report Excerpt	GPT-4 Turbo Label	Expert-Corrected Label
Pneumonia	“Stable mass surrounding the bronchi of the lower lobe of the left lung.”	1	0
Pulmonary Fibrosis	“... accompanied by sequela parenchymal changes in the lower lobe basal segments ...”	0	1
Spondylosis	“Degenerative changes are observed in the bone structures in the study area.”	0	1
Atherosclerosis	“Calibration of the aortic arch is at the maximal physiological limit. Calibration of other major vascular structures in the mediastinum is natural.”	1	0
Lung Mass	“There is a nodule measuring 7 mm in size, with the onset of cavitation in the center of the right lung.”	1	0

This human-in-the-loop verification framework ensures that the final CT-RATE-LT dataset preserves the scalability of GPT-4 Turbo auto-labeling while achieving expert-level accuracy and clinical reliability.

Prompting LLM for CT-RATE-LT Annotation

"You are a senior radiologist. Your task is to classify the given chest CT report into 15 abnormality categories.\n\nFor each category, output a binary label:\n- 1 if the condition is present\n- 0 if the condition is absent\n\nFollow these internal reasoning steps silently:\n1. Extract relevant evidence from the report.\n2. Normalize synonyms to standard medical terms.\n3. Resolve negations and uncertain expressions.\n4. Ensure logical consistency among labels.\n\nCategories to classify:\n1. Pneumonia\n2. Atherosclerosis\n3. Spondylosis\n4. Pneumothorax\n5. Bronchial thickening\n6. Osteophyte\n7. Empyema\n8. Lung cyst\n9. Lung mass\n10. Pericardial calcification\n11. Pneumomediastinum\n12. Pulmonary fibrosis\n13. Tuberculosis\n14. Pleural calcification\n15. Rib fracture\n\nReturn only a JSON object where each key is the category name and the value is either 0 or 1.\nStrictly follow this format without any extra commentary or explanation.\n\nExample:\n{\n \"Pneumonia\": 1,\n \"Atherosclerosis\": 0,\n \"Spondylosis\": 1,\n \"Pneumothorax\": 0,\n \"Bronchial thickening\": 1,\n \"Osteophyte\": 1,\n \"Empyema\": 0,\n \"Lung cyst\": 0,\n \"Lung mass\": 0,\n \"Pericardial calcification\": 0,\n \"Pneumomediastinum\": 0,\n \"Pulmonary fibrosis\": 1,\n \"Tuberculosis\": 0,\n \"Pleural calcification\": 0,\n \"Rib fracture\": 0\n}\n\nFindings:\n{Finding}\n\nImpression:\n{Impression}\n\nAssistant:"

Figure S1: Prompt template used for GPT-4 Turbo labeling of long-tail abnormalities in the CT-RATE-LT dataset.

Appendix B. Comprehensive Confusion Analysis Across Abnormality Categories

Unlike multi-class classification tasks where a conventional confusion matrix can be directly computed, our problem is a multi-label abnormality classification task in which each CT scan may contain multiple co-existing diseases. Under this setting, a standard confusion matrix becomes less informative, as the notion of a single “true vs. predicted” category does not apply. Therefore, we adopt an alternative formulation using **multi-label conditional confusion metrics** that more appropriately capture inter-class dependencies and misclassification patterns.

Specifically, Fig. S2 and Fig. S3 present comprehensive multi-label confusion heatmaps for the CT-RATE and RAD-ChestCT datasets, respectively. These heatmaps include:

- **Recall-oriented conditional confusion** $P(\text{pred} = 1 | \text{true} = 1)$, which quantifies how often a true abnormality tends to activate other predictions;
- **Precision-oriented conditional confusion** $P(\text{true} = 1 | \text{pred} = 1)$, which measures how often a predicted abnormality coincides with other true labels;
- **Co-activation rate** $P(\text{pred}_i = 1 \& \text{pred}_j = 1)$, reflecting joint activations between disease predictions;
- **Pearson correlation** of predicted scores, capturing continuous dependencies between abnormality likelihoods.

From the CT-RATE results (Fig. S2), several clear inter-class patterns emerge. High recall-oriented conditional confusion values are concentrated among fibrotic and bronchial categories, particularly between *Pulmonary fibrotic sequela*, *Peribronchial thickening*, and *Bronchiectasis*, suggesting that the model frequently co-activates these structurally related abnormalities. This

behavior aligns with clinical reality, as all three involve reticular or bronchial wall thickening patterns that often co-exist in chronic airway disease. Similarly, precision-oriented confusion indicates that predictions of *Pneumonia* often coincide with true positives of *Pulmonary fibrosis* or *Consolidation*, reflecting the difficulty of zero-shot models in disentangling acute from chronic opacities without fine-grained supervision. The co-activation map further reveals consistent clusters within pleural and parenchymal findings, such as *Pleural effusion* with *Lung opacity* and *Peribronchial thickening* with *Bronchiectasis*. These relationships correspond to anatomical proximity rather than label noise, thereby indicating that the model implicitly learns contextual co-occurrence structures among radiological regions.

For the RAD-ChestCT dataset (Fig. S3), the overall confusion magnitude is lower, reflecting higher category diversity and greater domain shift compared with CT-RATE. Nevertheless, stable correlation structures remain evident: *Atelectasis*, *Emphysema*, and *Lung opacity* exhibit mutual dependencies, while *Tuberculosis* and *Pulmonary fibrotic sequela* show strong linear correlations in Pearson analysis, consistent with post-infectious fibrotic remodeling patterns in real-world imaging. Moreover, co-activation patterns indicate that rare abnormalities such as *Pneumothorax* and *Pericardial effusion* remain largely independent, suggesting that the model does not produce spurious correlations on infrequent conditions, which represents a desirable property under zero-shot settings.

These findings collectively demonstrate that the proposed zero-shot model does not rely on accidental label correlations but captures clinically meaningful co-occurrence and feature dependencies. Most residual confusions occur between visually or pathologically overlapping entities, highlighting that model uncertainty primarily arises from radiological ambiguity rather than representational instability. Therefore, the conditional confusion framework provides not only a quantitative tool for measuring inter-class consistency but also qualitative evidence of the model’s capacity to generalize across semantically adjacent disease categories.

To further illustrate representative confusion sources, we visualize two disease pairs that exhibit strong inter-class correlation and high co-activation. As shown in Fig. S4, we analyze *Pulmonary fibrotic sequela* and *Interlobular septal thickening*. Panels (a) and (b) correspond to *Pulmonary fibrotic sequela* (success and failure cases, respectively), while (c) and (d) correspond to *Interlobular septal thickening*. Similarly, Fig. S5 examines confusion between *Peribronchial thickening* and *Bronchiectasis*. Each example displays the cosine similarity between image and text embeddings ($\cos(i, t)$) and corresponding attention heatmaps to highlight model focus regions.

These visualizations collectively demonstrate that confusion mainly arises from overlapping radiological patterns, such as reticular opacities, fibrotic streaks, or peribronchial structural changes, which remain challenging even for expert interpretation.

In summary, instead of a multi-class confusion matrix, the proposed multi-label conditional confusion analysis provides a more faithful and interpretable representation of inter-class ambiguity, thereby strengthening the robustness and clinical interpretability of our model evaluation.

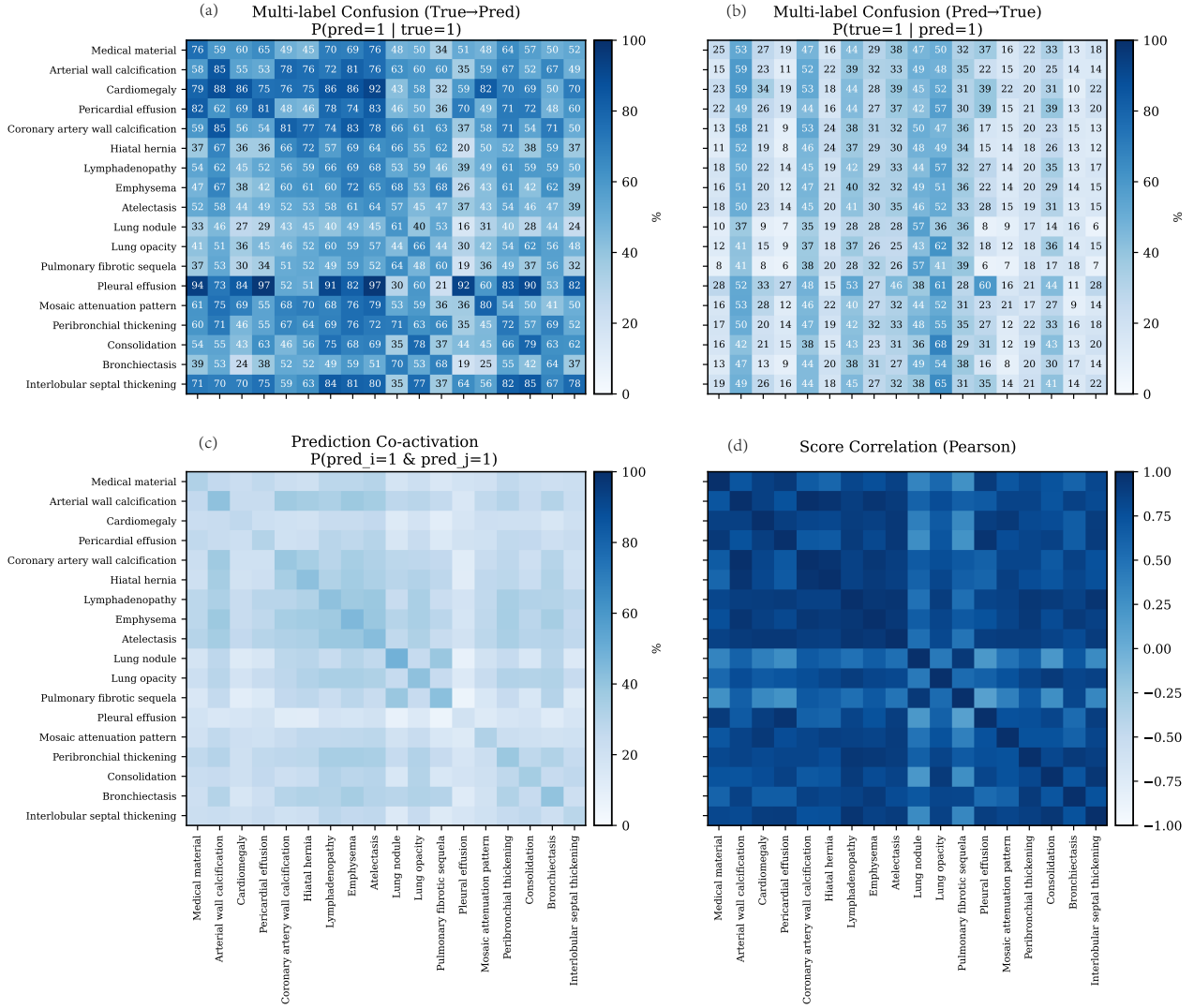


Figure S2: Comprehensive visualization of 18-class abnormality prediction on the CT-RATE dataset. (a) Recall-oriented conditional confusion $P(\text{pred} = 1 | \text{true} = 1)$. (b) Precision-oriented conditional confusion $P(\text{true} = 1 | \text{pred} = 1)$. (c) Co-activation rates $P(\text{pred}_i = 1 \& \text{pred}_j = 1)$. (d) Pearson correlation of predicted scores, revealing label co-occurrence and feature dependencies.

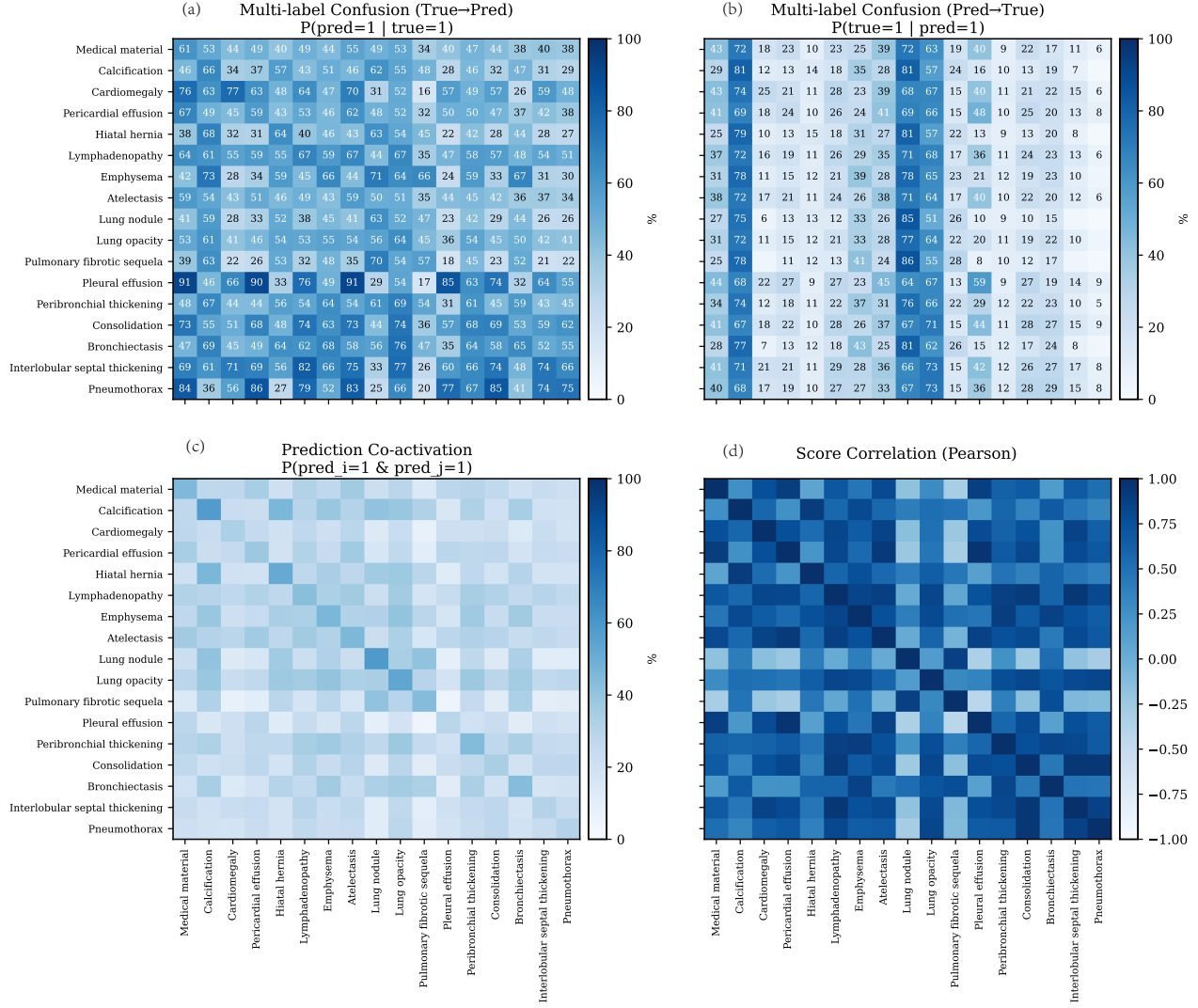


Figure S3: Comprehensive visualization of 17-class abnormality prediction on the RAD-ChestCT dataset. (a) Recall-oriented conditional confusion $P(\text{pred} = 1 | \text{true} = 1)$. (b) Precision-oriented conditional confusion $P(\text{true} = 1 | \text{pred} = 1)$. (c) Co-activation rates $P(\text{pred}_i = 1 \& \text{pred}_j = 1)$. (d) Pearson correlation of predicted scores, revealing label co-occurrence and feature dependencies.

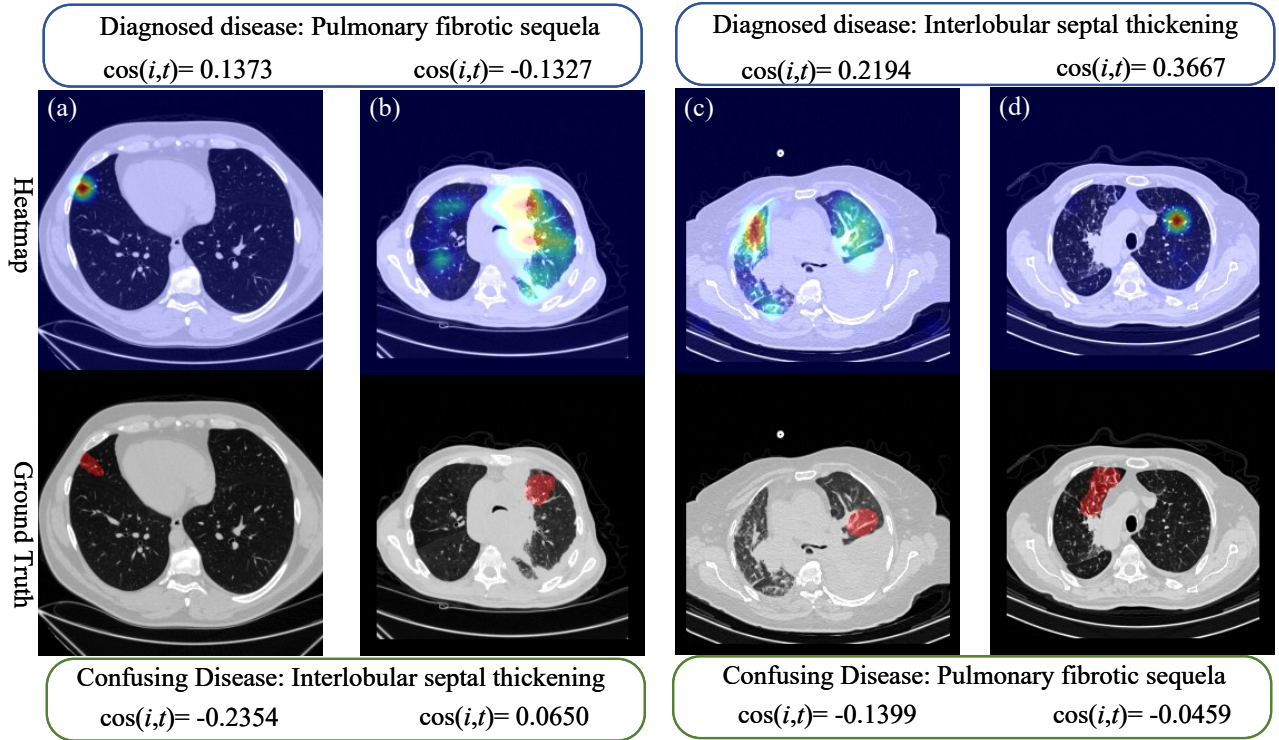


Figure S4: Representative visualization of confusion cases between **Pulmonary fibrotic sequela** and **Interlobular septal thickening**. Specifically, (a) and (b) correspond to the diagnosis of *Pulmonary fibrotic sequela*, where (a) shows a successful prediction and (b) illustrates a failure case. (c) and (d) correspond to *Interlobular septal thickening*, where (c) is a successful case and (d) is a failure case. Each sample presents the cosine similarity between the image and text embeddings ($\cos(i,t)$) and the corresponding attention heatmap indicating the key regions attended by the model.

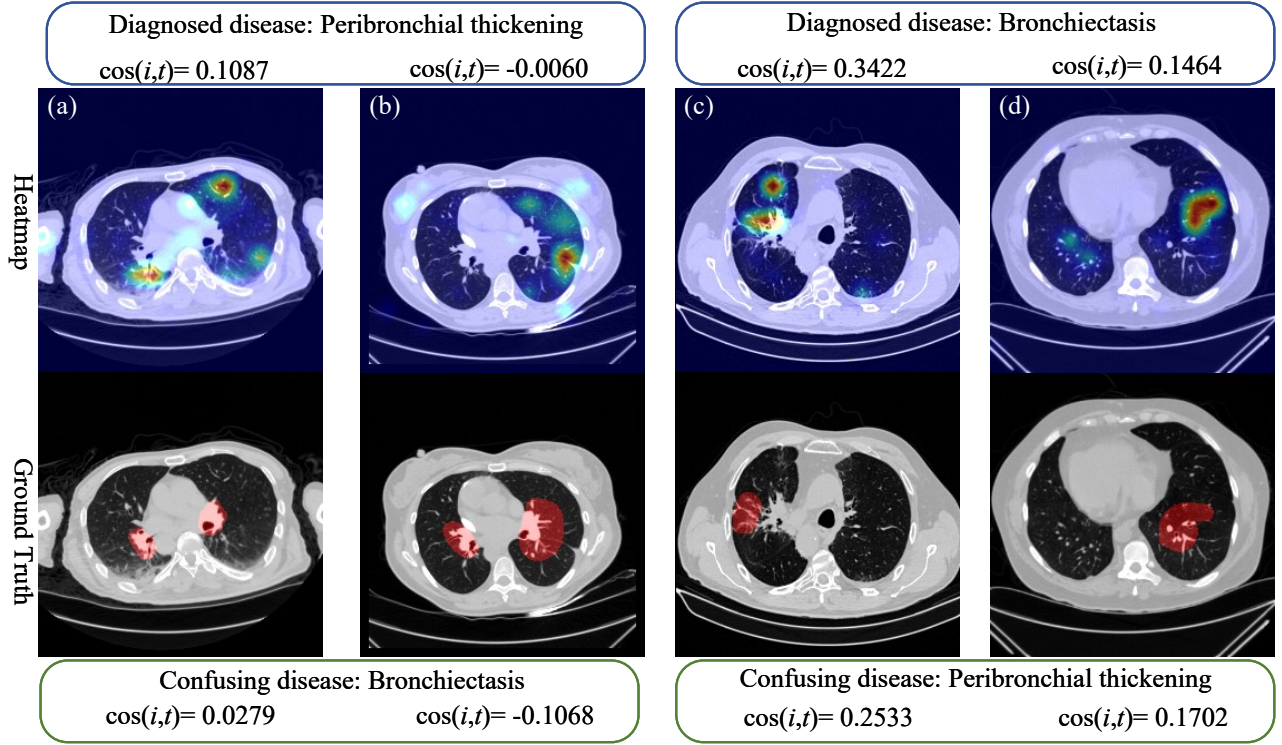


Figure S5: Representative visualization of confusion cases between **Peribronchial thickening** and **Bronchiectasis**. Specifically, (a) and (b) correspond to the diagnosis of *Peribronchial thickening*, where (a) shows a successful prediction and (b) illustrates a failure case. (c) and (d) correspond to *Bronchiectasis*, where (c) is a successful case and (d) is a failure case. Each sample presents the cosine similarity between the image and text embeddings ($\cos(i,t)$) and the corresponding attention heatmap indicating the key regions attended by the model.